```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load the image
image_path = '/Screenshot 2026-08-09 202635.png' # Corrected to use the ava
img = cv2.imread(image_path)

if img is None:
    print(f"Error: Could not load image from {image_path}. Please check the
else:
    # Resize the image to 500x500 to match the original example's dimensio
    # This ensures the homography points remain relevant
    img = cv2.resize(img, (500, 500))

    # ----------------------------------------------------
    # Source Points
    # ----------------------------------------------------
    src = np.array([
        [100,100],
        [400,100],
        [400,400],
        [100,400]
    ], dtype=np.float64)

    # ----------------------------------------------------
    # Destination Points
    # ----------------------------------------------------
    dst = np.array([
        [150,80],
        [420,120],
        [380,420],
        [80,350]
    ], dtype=np.float64)

    # ----------------------------------------------------
    # Direct Linear Transformation (DLT)
    # ----------------------------------------------------
    A = []

    for i in range(4):
        x, y = src[i]
        u, v = dst[i]

        A.append([-x,-y,-1,0,0,0,u*x,u*y,u])
        A.append([0,0,0,-x,-y,-1,v*x,v*y,v])

    A = np.array(A)

    # Solve using Singular Value Decomposition (SVD)
    U, S, VT = np.linalg.svd(A)

    # Homography Matrix
```

```
    H = VT[-1].reshape(3,3)

    # Normalize
    H = H / H[2,2]

    print("Homography Matrix (DLT):")
    print(H)

    # --------------------------------------------------
    # Apply Transformation
    # --------------------------------------------------
    warped = cv2.warpPerspective(img, H, (500,500))

    # Convert BGR to RGB for display
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    warped_rgb = cv2.cvtColor(warped, cv2.COLOR_BGR2RGB)

    # --------------------------------------------------
    # Display
    # --------------------------------------------------
    plt.figure(figsize=(12,6))

    plt.subplot(1,2,1)
    plt.imshow(img_rgb)
    plt.title("Input Image")
    plt.axis("off")

    plt.subplot(1,2,2)
    plt.imshow(warped_rgb)
    plt.title("DLT Transformed Image")
    plt.axis("off")

    plt.show()
```

```
Homography Matrix (DLT):
[[ 7.49847716e-01 -2.47445008e-01  9.10795262e+01]
 [ 9.75972927e-02  7.27106599e-01 -7.09983080e+00]
 [-2.33502538e-04 -3.45177665e-04  1.00000000e+00]]
```

Input Image

DLT Transformed Image